Chapter 5: Sturm–Liouville Theory

Special Functions of Mathematical Physics: A Tourist's Guidebook

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Outline

1. Review of linear algebra
2. Sturm–Liouville operators

Where we are going

A short chapter on the *theory* behind the equations we have been solving.

- The Frobenius method gave us series solutions of second-order linear ODEs. Now we ask what those solutions have in common.
- Many of them are eigenfunctions of a *Sturm–Liouville equation*

$$\frac{d}{dx}\left(p(x)\frac{du}{dx}\right) + q(x)u(x) = -\lambda\, w(x)\, u(x),$$

  which arises from separation of variables in countless physical problems.
- We first review **function spaces**: vectors, inner products, norms, orthogonality, and the operators that act on them.
- We then identify when a second-order operator is **self-adjoint**, prove its eigenfunctions are **orthogonal**, and learn to put any such operator into Sturm–Liouville form.
- The payoff: orthogonal families like the **Legendre polynomials** fall out automatically.

Table of contents

1. Review of linear algebra
2. Sturm–Liouville operators

Vector spaces

A *vector space* is a collection of objects (called vectors) that is closed under addition and scalar multiplication: any linear combination of vectors in the collection also lies in the collection. The objects need not be arrows—we will primarily be interested in **functions as vectors**. If $u(x)$ and $v(x)$ are both vectors in a given space, then

$$a\, u(x) + b\, v(x)$$

is also in that vector space, for any scalars a and b.

Examples.

- The polynomials of degree at most two, $a_0 + a_1 x + a_2 x^2$: scale one or add two together and the result still has degree at most two.
- The smooth functions on $[0, 1]$ that vanish at the endpoints: linear combinations still vanish at the endpoints.

Vector spaces (cont.)

A *basis* for a vector space is a linearly independent spanning set: every vector is built from it, and no basis vector is redundant. For the quadratics, the set $\{1, x, x^2\}$ is a basis:

$$1 + 2x + 3x^2 = 1 \cdot 1 + 2 \cdot x + 3 \cdot x^2. \tag{5.1.1}$$

The set $\{1, 2x - 1, 6x^2 - 6x + 1\}$ is *another* basis for the same space. Building the same polynomial in this basis,

$$1 + 2x + 3x^2 = 3 \cdot 1 + \tfrac{5}{2}\,(2x - 1) + \tfrac{1}{2}\,(6x^2 - 6x + 1). \tag{5.1.2}$$

(Check the x^2 term: $\frac{1}{2} \cdot 6 = 3$; the x term: $\frac{5}{2} \cdot 2 + \frac{1}{2} \cdot (-6) = 5 - 3 = 2$; the constant: $3 - \frac{5}{2} + \frac{1}{2} = 1$.)

There are infinitely many bases. The number of vectors in a finite basis is the *dimension* of the space; the quadratics have dimension three. Some spaces have no finite basis. A *Schauder basis* is a sequence (e_n) such that every vector has a unique norm-convergent expansion $\sum_n a_n e_n$. For our purposes, the important engineering example is that

Vector spaces (cont.)

$\{\sin \pi x, \sin 2\pi x, \sin 3\pi x, \dots\}$ is an orthogonal basis for $L^2(0,1)$, and smooth functions on $[0,1]$ that vanish at the endpoints have sine-series representations of the form

$$f(x) = \sum_{n=1}^{\infty} a_n \sin n\pi x. \tag{5.1.3}$$

Inner product spaces

An *inner product*, written $(\cdot,\cdot)$, maps two vectors to a complex number and has three properties: linearity, conjugate symmetry (Hermiticity), and positive-definiteness.

1. $(au + bv,\ w) = a\,(u, w) + b\,(v, w)$,
2. $(u, v) = (v, u)^*$,
3. $(u, u) \geq 0$, and $(u, u) = 0$ if and only if $u = 0$,

where u, v, w are vectors and a, b are scalar values. Combining conjugate symmetry with linearity gives the conjugate-linearity in the second slot, $(u, av) = a^*(u, v)$.

The common choice is the L^2*-inner product*

$$(u, v) = \int_a^b u(x)\,v(x)^*\,dx, \tag{5.1.4}$$

where v^* is the complex conjugate of v and a, b are boundary points (possibly $\pm\infty$). A useful generalization is the *weighted* inner product

$$(u, v)_w = \int_a^b u(x)\,v(x)^*\,w(x)\,dx \tag{5.1.5}$$

Inner product spaces (cont.)

for a real weighting function $w(x) > 0$ except possibly at isolated endpoints. Positivity of w is what keeps $(u, u)_w$ positive for every nonzero function. A vector space carrying an inner product is an *inner product space*.

Inner product spaces (cont.)

A *norm* $\|\cdot\|$ is a real-valued function with three properties:

1. $\|u\| \geq 0$, and $\|u\| = 0$ if and only if $u \equiv 0$ (positive-definiteness),
2. $\|au\| = |a|\,\|u\|$ (absolute homogeneity),
3. $\|u+v\| \leq \|u\| + \|v\|$ (triangle inequality / subadditivity).

An inner product induces a natural norm $\|u\| = \sqrt{(u,u)}$, and similarly a weighted norm $\|u\|_w = \sqrt{(u,u)_w}$. We *normalize* a function u by dividing it by its norm.

Two functions are *orthogonal* when their inner product is zero, $(u,v) = 0$. An *orthonormal* basis $\{u_1, \ldots, u_n\}$ is one whose elements are mutually orthogonal and have unit norm:

$$(u_i, u_j) = \delta_{ij}, \tag{5.1.6}$$

with the Kronecker delta $\delta_{ii} = 1$ and $\delta_{ij} = 0$ for $i \neq j$.

The great benefit: in an orthonormal basis, the coefficients of a decomposition $f = \sum_{i=1}^{n} a_i u_i$ are read off by a single inner product,

$$(f, u_j) = \left(\sum_{i=1}^{n} a_i u_i,\ u_j\right) = \sum_{i=1}^{n} a_i\,(u_i, u_j) = \sum_{i=1}^{n} a_i\,\delta_{ij} = a_j. \tag{5.1.7}$$

Inner product spaces (cont.)

So $a_j = (f, u_j)$.

Example: orthogonality of $\sin m\pi x$ and $\sin n\pi x$

Show that $\sin m\pi x$ and $\sin n\pi x$ (m, n positive integers) are orthogonal in the L^2-inner product over $[0, 1]$. First an identity for the integrand. Writing $\sin nx = (e^{inx} - e^{-inx})/2i$,

$$\begin{aligned}
\sin mx \sin nx &= \operatorname{Im}\big(e^{imx} \sin nx\big) \\
&= \operatorname{Im}\left(e^{imx}\, \frac{e^{inx} - e^{-inx}}{2i}\right) \\
&= \operatorname{Im}\left(\frac{e^{i(m+n)x} - e^{i(m-n)x}}{2i}\right) \\
&= \tfrac{1}{2} \operatorname{Re}\big(e^{i(m-n)x} - e^{i(m+n)x}\big) \\
&= \tfrac{1}{2}\big(\cos(m-n)x - \cos(m+n)x\big). \qquad (5.1.8)
\end{aligned}$$

(Here $\operatorname{Im}(w/i) = -\operatorname{Re}(w)$, since dividing by i rotates by -90°; the extra minus sign swaps the order of the two cosines.)

Example: orthogonality of $\sin m\pi x$ and $\sin n\pi x$ (cont.)

Therefore, when $m \neq n$,

$$\int_0^1 \sin m\pi x \, \sin n\pi x \, dx = \left[\frac{\sin(m-n)\pi x}{2(m-n)\pi} - \frac{\sin(m+n)\pi x}{2(m+n)\pi} \right]_0^1 = 0, \tag{5.1.9}$$

since $\sin(k\pi) = 0$ for every integer k. When $m = n$,

$$\int_0^1 \sin^2 m\pi x \, dx = \left[\frac{x}{2} - \frac{\sin 2m\pi x}{4m\pi} \right]_0^1 = \frac{1}{2}. \tag{5.1.10}$$

Collecting both cases,

$$(\sin m\pi x, \, \sin n\pi x) = \tfrac{1}{2} \, \delta_{mn}. \tag{5.1.11}$$

The sines are orthogonal; their norm is $1/\sqrt{2}$, not 1.

Example: the sine transform

Initial conditions $f(x)$ for the guitar string are represented by a sine series over $[0, 1]$,

$$f(x) = \sum_{n=1}^{\infty} a_n \sin n\pi x. \tag{5.1.12}$$

Because $\{\sin \pi x, \sin 2\pi x, \sin 3\pi x, \dots\}$ is an orthogonal basis, the coefficient formula (5.1.7) (adjusted for the norm $\|u_n\|^2 = 1/2$ from (5.1.11)) gives

$$a_n = \frac{(f, u_n)}{\|u_n\|^2} = \frac{1}{1/2} \int_0^1 f(x) \sin n\pi x \, dx = 2 \int_0^1 f(x) \sin n\pi x \, dx. \tag{5.1.13}$$

This is the *sine transform* of $f(x)$: orthogonality turns the search for coefficients into a single integral per mode.

Linear operators and the adjoint

A *linear operator* L satisfies $\mathrm{L}(a\,u) = a\,\mathrm{L}u$ and $\mathrm{L}(u+v) = \mathrm{L}u + \mathrm{L}v$. For example the derivative $\frac{d}{dx}$, the second-order operator $(x+1)\frac{d^2}{dx^2}$, and the antiderivative are all linear.

Given two linear operators L and $\tilde{\mathrm{L}}$ on an inner product space, $\tilde{\mathrm{L}}$ is the *adjoint* of L if

$$(u, \mathrm{L}v) = (\tilde{\mathrm{L}}u,\ v) \quad \text{for all } u, v. \tag{5.1.14}$$

Take $\mathrm{L} = \frac{d}{dx}$ on the smooth functions vanishing at 0 and 1. In the integration-by-parts computations below we use real-valued functions; the complex version is the same calculation with conjugates inserted according to (5.1.4). Integration by parts gives

$$(u, \mathrm{L}v) = \int_0^1 u\,\frac{dv}{dx}\,dx = [uv]_0^1 - \int_0^1 \frac{du}{dx}\,v\,dx = -\int_0^1 \frac{du}{dx}\,v\,dx = (-\mathrm{L}u,\ v), \tag{5.1.15}$$

the boundary term vanishing because u, v are zero at 0 and 1. So the adjoint of $\frac{d}{dx}$ is $-\frac{d}{dx}$.

Linear operators and the adjoint (cont.)

When $(u, \mathrm{L}v) = (\mathrm{L}u,\ v)$ for all u, v—that is, when L is its own adjoint—we call L *self-adjoint* or *Hermitian*. The first derivative is *not* self-adjoint (its adjoint carries a minus sign). But the second derivative $\frac{d^2}{dx^2}$ *is* self-adjoint on the smooth functions vanishing at 0 and 1. If $u(0) = u(1) = v(0) = v(1) = 0$, then

$$(u, v'') = [uv']_0^1 - \int_0^1 u'v'\,dx = -[u'v]_0^1 + \int_0^1 u''v\,dx = (u'', v).$$

The two boundary terms vanish for Dirichlet functions, and the two integrations by parts return the derivative to the first slot without changing the sign.

Eigenfunctions and the orthogonality theorem

The *eigenfunctions* of an operator L are functions that L merely scales. The scale factor is the *eigenvalue*. The eigenvalue problem

$$\mathrm{L}u(x) = \lambda\, u(x) \tag{5.1.16}$$

asks for the pair (u, λ). The *generalized* eigenvalue problem inserts a weight,

$$\mathrm{L}u(x) = \lambda\, w(x)\, u(x). \tag{5.1.17}$$

In the Sturm–Liouville convention below we instead write $\mathrm{L}u = -\lambda wu$, so that common examples have positive λ. The prefix *eigen* means "own" or "characteristic": an eigenfunction keeps its own shape under the operator, changing only by a scalar multiplier (or, in the weighted problem, by a scalar times the prescribed weight).

Example. On a space that contains them, the derivative $\frac{d}{dx}$ has eigenfunctions $e^{\lambda x}$ with eigenvalue λ. The functions $\cos mx$ and $\sin mx$ are eigenfunctions of $\frac{d^2}{dx^2}$ with eigenvalue $-m^2$. The function space matters: on polynomials of degree at most 2, $\frac{d}{dx}$ has only constant eigenfunctions, with eigenvalue 0.

Eigenfunctions and the orthogonality theorem (cont.)

Theorem

The eigenvalues of a self-adjoint operator are real, and eigenfunctions belonging to distinct eigenvalues are orthogonal.

Proof. Let u_i and u_j be eigenfunctions with eigenvalues λ_i and λ_j. Because L is self-adjoint, $(\mathrm{L}u_i, u_j) = (u_i, \mathrm{L}u_j)$, so

$$\begin{aligned} 0 &= (\mathrm{L}u_i, u_j) - (u_i, \mathrm{L}u_j) \\ &= (\lambda_i u_i, u_j) - (u_i, \lambda_j u_j) \\ &= \left(\lambda_i - \lambda_j^*\right)(u_i, u_j). \end{aligned} \tag{5.1.18}$$

(The conjugate λ_j^* appears because the scalar in the second slot pulls out conjugated.) For $i = j$ the inner product $(u_i, u_i) > 0$, so the prefactor must vanish: $\lambda_i = \lambda_i^*$, i.e. the eigenvalues are **real**. Now that each eigenvalue is real, $\lambda_j^* = \lambda_j$, so the prefactor for $i \neq j$ is the real number $\lambda_i - \lambda_j$. Since $\lambda_i \neq \lambda_j$ this prefactor is nonzero, and (5.1.18) forces $(u_i, u_j) = 0$: the eigenfunctions are **orthogonal**. If several independent eigenfunctions share the same eigenvalue, orthogonality is not automatic, but Gram–Schmidt can choose an orthogonal basis inside that eigenspace. □

Eigenfunctions and the orthogonality theorem (cont.)

Application. The harmonic-oscillator equation

$$\frac{d^2u}{dx^2} + (m\pi)^2 u = 0$$

is $\mathrm{L}u = -(m\pi)^2 u$ with $\mathrm{L} = \frac{d^2}{dx^2}$. The operator L is self-adjoint in the L^2-space over $[0, 1]$, so by the theorem its eigenfunctions $\sin m\pi x$ are mutually orthogonal—recovering (5.1.11) *without* the integral computation.

Table of contents

1. Review of linear algebra
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When is a second-order operator self-adjoint?

Consider the general linear second-order operator

$$\mathrm{L}u = p_0 \frac{d^2 u}{dx^2} + p_1 \frac{du}{dx} + p_2 u, \tag{5.2.1}$$

with real-valued coefficients p_0, p_1, p_2, acting on the real L^2-space of smooth functions that vanish at $x = a$ and $x = b$. We find the adjoint by taking the inner product against v and integrating by parts. For complex function spaces, the same boundary-term calculation is used after inserting the conjugates from (5.1.4).

$$(v, \mathrm{L}u) = \int_a^b \left(v p_0 u'' + v p_1 u' + v p_2 u \right) dx. \tag{5.2.2}$$

Integrate by parts term by term. For the first term, one pass moves a derivative off u'', a second pass moves it off u', each pass flipping a sign and leaving a boundary piece:

$$\begin{aligned} \int_a^b (v p_0) u'' \, dx &= (v p_0) u' \Big|_a^b - \int_a^b (v p_0)' u' \, dx \\ &= \Big[(v p_0) u' - (v p_0)' u \Big]_a^b + \int_a^b (v p_0)'' u \, dx. \end{aligned}$$

When is a second-order operator self-adjoint? (cont.)

For the second term, one pass suffices:

$$\int_a^b (vp_1)u'\,dx = (vp_1)u\Big|_a^b - \int_a^b (vp_1)'u\,dx.$$

The third term, $\int (vp_2)u\,dx$, needs no parts. Collecting the three boundary pieces and the three integrals,

$$(v, \mathrm{L}u) = \underbrace{\Big[-(vp_0)'u + (vp_0)u' + (vp_1)u\Big]_a^b}_{\text{boundary terms}} + \int_a^b \big((vp_0)''u - (vp_1)'u + (vp_2)u\big)\,dx. \tag{5.2.3}$$

When is a second-order operator self-adjoint? (cont.)

Both u and v vanish at $x = a, b$, so the boundary terms drop, leaving

$$(v, \mathrm{L}u) = \int_a^b \big((vp_0)'' - (vp_1)' + (vp_2)\big)\, u\, dx = (\mathrm{L}^* v,\ u). \tag{5.2.4}$$

So the adjoint is

$$\mathrm{L}^* u = \frac{d^2}{dx^2}(p_0 u) - \frac{d}{dx}(p_1 u) + p_2 u. \tag{5.2.5}$$

The operator is self-adjoint when $\mathrm{L} = \mathrm{L}^*$. Expanding the adjoint,

$$\begin{aligned} \mathrm{L}^* u &= (p_0 u)'' - (p_1 u)' + p_2 u \\ &= p_0'' u + 2p_0' u' + p_0 u'' - p_1' u - p_1 u' + p_2 u. \end{aligned} \tag{5.2.6}$$

When is a second-order operator self-adjoint? (cont.)

Subtract $\mathrm{L}u = p_0 u'' + p_1 u' + p_2 u$ from (5.2.6). The u'' and $p_2 u$ terms cancel, and after canceling and rearranging we are left with

$$p_0''\, u + 2p_0'\, u' - 2p_1\, u' - p_1'\, u = 0. \tag{5.2.7}$$

Grouping the u' terms gives $2(p_0' - p_1)u'$ and the u terms give $(p_0'' - p_1')u = (p_0' - p_1)'u$, so the condition reads

$$2(p_0' - p_1)\, u' + (p_0' - p_1)'\, u = 0.$$

This must hold for *every* u in the space, not just one. Writing $r = p_0' - p_1$, the requirement $2r\, u' + r'\, u = 0$ for all u forces the coefficient of each independent piece to vanish: $r = 0$ kills the u' term, and then $r' = 0$ kills the u term automatically. Hence

$$p_0' = p_1. \tag{5.2.8}$$

This is the condition for self-adjointness of (5.2.1).

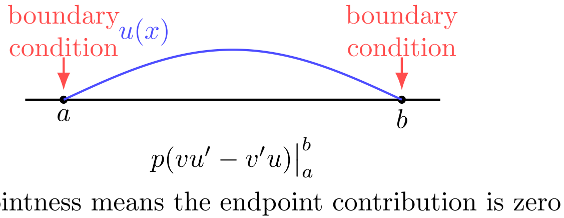
When $p_0' = p_1$, the operator (5.2.1) collapses to a perfect derivative. Indeed $p_0 u'' + p_0' u' = (p_0 u')'$, so relabeling $p(x) = p_0(x)$ and $q(x) = p_2(x)$,

$$\mathrm{L}u = \frac{d}{dx}\left(p(x)\frac{du}{dx}\right) + q(x)\, u. \tag{5.2.9}$$

This is the *Sturm–Liouville operator*.

With $p_0' = p_1$ the boundary terms of (5.2.3) simplify to

$$p_0\,(vu' - v'u)\Big|_a^b, \tag{5.2.10}$$



The Sturm–Liouville operator (cont.)

which vanishes under any of four boundary conditions:

1. $p(a) = p(b) = 0$ (the coefficient itself vanishes at the ends);
2. $u(a) = u(b) = v(a) = v(b) = 0$ (Dirichlet—values vanish);
3. $u'(a) = u'(b) = v'(a) = v'(b) = 0$ (Neumann—derivatives vanish);
4. $p(a) = p(b)$ and $u(a) = u(b),\ u'(a) = u'(b),\ v(a) = v(b),\ v'(a) = v'(b)$ (periodic).

The Sturm–Liouville operator (cont.)

So the Sturm–Liouville operator is self-adjoint whenever the geometry supplies one of these four conditions. The weight $w(x)$ in the equation below does not by itself cancel the boundary term; it enters the orthogonality integral. On infinite intervals, boundary terms are controlled by the coefficient $p(x)$ and by the decay of the eigenfunctions. For example, Hermite polynomial modes satisfy

$$e^{-x^2}\big(vu' - v'u\big) \longrightarrow 0 \qquad (x \to \pm\infty),$$

because the exponential factor dominates the polynomial growth.

When L is a Sturm–Liouville operator, the generalized eigenvalue equation $\mathrm{L}u = -\lambda\, w\, u$ is the *Sturm–Liouville equation*

$$\frac{d}{dx}\left(p(x)\frac{du}{dx}\right) + q(x)\,u(x) = -\lambda\, w(x)\, u(x), \tag{5.2.11}$$

First check that the generalized eigenvalues are real. Setting $i = j$ in the subtraction below,

$$0 = (\mathrm{L}u_i, u_i) - (u_i, \mathrm{L}u_i) = -(\lambda_i - \lambda_i^*)\int_a^b |u_i(x)|^2 w(x)\, dx,$$

The Sturm–Liouville operator (cont.)

and the weighted norm is positive, so $\lambda_i = \lambda_i^*$. For two solutions u_i, u_j with distinct real λ_i, λ_j, self-adjointness gives

$$(\mathrm{L}u_i, u_j) - (u_i, \mathrm{L}u_j) = -(\lambda_i - \lambda_j) \int_a^b u_i(x) u_j(x)^* w(x)\, dx = 0.$$

Thus the eigenfunctions are orthogonal in the weighted inner product.

Example: Legendre and Bessel in self-adjoint form

Legendre. Put the Legendre equation

$$(1-x^2)y'' - 2xy' + m(m+1)y = 0$$

into self-adjoint form. Since $-2x$ is exactly the derivative of $1-x^2$, the first two terms combine into a single derivative:

$$\frac{d}{dx}\left((1-x^2)\frac{dy}{dx}\right) = -\,m(m+1)\,y. \tag{5.2.12}$$

Comparing with $\mathrm{L}y = -\lambda w y$: the eigenvalues are $\lambda = +m(m+1)$ and the weight is $w(x) = 1$. (Here $p(x) = 1 - x^2$, $q(x) = 0$.)

Example: Legendre and Bessel in self-adjoint form (cont.)

Bessel / circular drum. For a unit-radius drum the Bessel function $J_m(x)$ solves

$$x^2 y'' + xy' + (x^2 - m^2)y = 0.$$

Rescale by $x = \alpha s$, so $\frac{d}{dx} = \frac{1}{\alpha}\frac{d}{ds}$ and $\frac{d^2}{dx^2} = \frac{1}{\alpha^2}\frac{d^2}{ds^2}$. Substituting $x^2 = \alpha^2 s^2$ into each term,

$$\alpha^2 s^2 \cdot \frac{1}{\alpha^2}\frac{d^2 y}{ds^2} + \alpha s \cdot \frac{1}{\alpha}\frac{dy}{ds} + (\alpha^2 s^2 - m^2)\, y = 0,$$

and the α's cancel in the first two terms, leaving

$$s^2 \frac{d^2 y}{ds^2} + s\frac{dy}{ds} + (\alpha^2 s^2 - m^2)\, y = 0. \tag{5.2.13}$$

Divide (5.2.13) by s, splitting $(\alpha^2 s^2 - m^2)/s = \alpha^2 s - m^2/s$:

$$s\frac{d^2 y}{ds^2} + \frac{dy}{ds} + \left(\alpha^2 s - \frac{m^2}{s}\right) y = 0.$$

Recognizing $sy'' + y' = (sy')'$ and moving the $\alpha^2 s\, y$ term to the right, the Sturm–Liouville form is

$$\frac{d}{ds}\left(s\frac{dy}{ds}\right) - \frac{m^2}{s}\, y = -\,\alpha^2\, s\, y. \tag{5.2.14}$$

Here $p(s) = s$, $q(s) = -m^2/s$, weight $w(s) = s$, eigenvalue $\lambda = \alpha^2$.

Example: Legendre and Bessel in self-adjoint form (cont.)

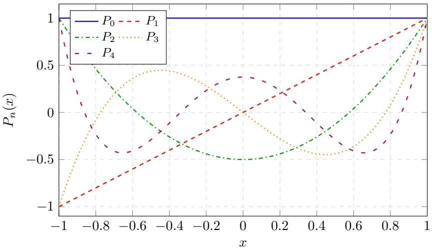
For fixed m, consider the eigenfunctions $J_m(\alpha_i s)$. Use one endpoint condition for the whole family. At $s = 1$ choose either $J_m(\alpha_i) = 0$ for every i or $J_m'(\alpha_i) = 0$ for every i; then the boundary term vanishes. At $s = 0$, the regular modes satisfy $J_m(\alpha s) = O(s^m)$ and $J_m'(\alpha s) = O(s^{m-1})$ (with $J_0'(\alpha s) = O(s)$), so

$$s\left(y_i y_j' - y_j y_i'\right) \to 0.$$

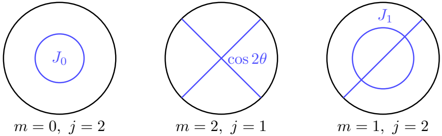
The singular Y_m modes are excluded by regularity at the center. Choosing the α_i to be the zeros of J_m, $J_m(\alpha_i) = 0$, the radial modes are mutually orthogonal with weight $w(s) = s$:

$$\int_0^1 J_m(\alpha_i s)\, J_m(\alpha_j s)\, s\, ds = 0 \quad \text{when } i \neq j. \tag{5.2.15}$$

Legendre and Bessel modes



Legendre modes are orthogonal on $[-1, 1]$ with weight $w(x) = 1$.



Drum modes use the radial weight $w(s) = s$; zeros at $s = 1$ select the allowed α_i.

Any second-order operator into Sturm–Liouville form

Even when $p_0' \neq p_1$, we can *force* an operator into self-adjoint form with an integrating factor on any interval where p_0 is nonzero and p_1/p_0 is integrable. Start from the general equation

$$p_0(x)\,\frac{d^2u}{dx^2} + p_1(x)\,\frac{du}{dx} + p_2(x)\,u = 0, \tag{5.2.16}$$

and divide by p_0:

$$\frac{d^2u}{dx^2} + \frac{p_1(x)}{p_0(x)}\,\frac{du}{dx} + \frac{p_2(x)}{p_0(x)}\,u = 0. \tag{5.2.17}$$

Multiply by the integrating factor

$$Q(x) = \exp\left(\int_a^x \frac{p_1(s)}{p_0(s)}\,ds\right), \qquad \text{so} \qquad Q' = \frac{p_1}{p_0}\,Q. \tag{5.2.18}$$

Any second-order operator into Sturm–Liouville form (cont.)

The first two terms then combine into a single derivative, because $Qu'' + Q'u' = (Qu')'$:

$$\frac{d}{dx}\left(Q(x)\,\frac{du}{dx}\right) + Q(x)\,\frac{p_2(x)}{p_0(x)}\,u = 0. \tag{5.2.19}$$

This is now Sturm–Liouville form with $p = Q$ and $q = Q\,p_2/p_0$.

Example (Hermite). Consider $y'' - 2xy' + \lambda y = 0$. Here $p_1/p_0 = -2x$, so the integrating factor is

$$Q(x) = \exp\left(\int_0^x -2s\,ds\right) = \exp\!\left(-x^2\right) = e^{-x^2}.$$

(Any finite base point works, since changing it only rescales Q by a constant, which does not affect self-adjoint form; we take 0 because $\int_{-\infty}^{x} -2s\,ds$ would diverge.) Multiplying through,

$$e^{-x^2}\left[y'' - 2xy' + \lambda y\right] = e^{-x^2}y'' - 2x\,e^{-x^2}y' + \lambda\,e^{-x^2}y = \left(e^{-x^2}y'\right)' + \lambda\,e^{-x^2}y, \tag{5.2.20}$$

Any second-order operator into Sturm–Liouville form (cont.)

since $\frac{d}{dx}e^{-x^2} = -2x\, e^{-x^2}$. The Hermite equation in self-adjoint form is therefore

$$\frac{d}{dx}\left(e^{-x^2}\frac{dy}{dx}\right) = -\,\lambda\, e^{-x^2}\, y, \tag{5.2.21}$$

with weight $w(x) = e^{-x^2}$. For Hermite polynomial modes, the endpoint boundary term vanishes because e^{-x^2} times the polynomial Wronskian tends to zero at $\pm\infty$.

Mathematical detour: the Gram–Schmidt process

Gram–Schmidt turns any set $\{u_1, u_2, u_3, \dots\}$ into a mutually orthogonal set $\{v_1, v_2, v_3, \dots\}$ by projecting out the components already accounted for. Begin with

$$v_1 = u_1. \tag{5.2.22}$$

Define v_2 as the part of u_2 orthogonal to v_1, obtained by subtracting the projection of u_2 onto v_1:

$$v_2 = u_2 - \left(u_2, \frac{v_1}{\|v_1\|} \right) \frac{v_1}{\|v_1\|} = u_2 - v_1 \frac{(u_2, v_1)}{\|v_1\|^2}. \tag{5.2.23}$$

Continue, projecting out the v_1 and v_2 components of u_3:

$$v_3 = u_3 - v_1 \frac{(u_3, v_1)}{\|v_1\|^2} - v_2 \frac{(u_3, v_2)}{\|v_2\|^2}. \tag{5.2.24}$$

Note the denominator of the last term is $\|v_2\|^2$ (the vector being projected onto), and the process continues for the remaining basis functions.

Mathematical detour: the Gram–Schmidt process (cont.)

Legendre polynomials from Gram–Schmidt. The Legendre equation can be solved by Frobenius; a second route is Gram–Schmidt. In Sturm–Liouville form $(pu')' + qu = -\lambda wu$ the Legendre equation reads

$$\big((1-x^2)y'\big)' = -\,m(m+1)\,y,$$

so $p(x) = 1 - x^2$, weight $w(x) = 1$, and $\lambda = m(m+1)$. If instead we view $L[y] = ((1-x^2)y')'$ as a plain operator, then its operator eigenvalue is $-m(m+1)$. Since $p(1) = p(-1) = 0$ (boundary condition 1), the operator is self-adjoint with respect to

$$(u, v) = \int_{-1}^{1} u\,v\,dx, \tag{5.2.25}$$

and its eigenfunctions are orthogonal in this inner product.

Mathematical detour: the Gram–Schmidt process (cont.)

Apply Gram–Schmidt to the monomials $\{1, x, x^2, x^3, \dots\}$ over $[-1, 1]$. Let $v_0, v_1, v_2, \dots$ denote the raw orthogonal polynomials before the final normalization. Start with $v_0(x) = 1$. For degree one, subtract the v_0 component from x:

$$v_1(x) = x - v_0(x)\,\frac{(v_0, x)}{\|v_0\|^2} = x - 1 \cdot \frac{(1, x)}{\|1\|^2} = x, \tag{5.2.26}$$

because $(1, x) = \int_{-1}^{1} x\,dx = 0$.

For degree two, subtract the v_0 and v_1 components from x^2:

$$\begin{aligned} v_2(x) &= x^2 - v_0(x)\,\frac{(v_0, x^2)}{\|v_0\|^2} - v_1(x)\,\frac{(v_1, x^2)}{\|v_1\|^2} \\ &= x^2 - 1 \cdot \frac{(1, x^2)}{\|1\|^2} - x \cdot \frac{(x, x^2)}{\|x\|^2}. \end{aligned} \tag{5.2.27}$$

Now $(1, x^2) = \int_{-1}^{1} x^2\,dx = \frac{2}{3}$ and $\|1\|^2 = \int_{-1}^{1} 1\,dx = 2$, so the second term is $\frac{2/3}{2} = \frac{1}{3}$; and $(x, x^2) = \int_{-1}^{1} x^3\,dx = 0$. Hence $v_2(x) = x^2 - \frac{1}{3}$. The standard Legendre polynomial uses the scaling $P_n(1) = 1$, so $P_0 = v_0$, $P_1 = v_1$, and $P_2 = \frac{3}{2} v_2 = \frac{1}{2}(3x^2 - 1)$.

Mathematical detour: the Gram–Schmidt process (cont.)

One more step shows what the bookkeeping looks like. Project the previous components out of x^3:

$$v_3(x) = x^3 - v_0 \frac{(v_0, x^3)}{\|v_0\|^2} - v_1 \frac{(v_1, x^3)}{\|v_1\|^2} - v_2 \frac{(v_2, x^3)}{\|v_2\|^2}.$$

The first numerator is $\int_{-1}^{1} x^3\,dx = 0$. The last numerator is also zero by oddness:

$$(v_2, x^3) = \int_{-1}^{1} \left(x^2 - \tfrac{1}{3}\right) x^3\,dx = \int_{-1}^{1} \left(x^5 - \tfrac{1}{3}x^3\right)\,dx = 0.$$

For the middle projection,

$$(v_1, x^3) = \int_{-1}^{1} x^4\,dx = \frac{2}{5}, \qquad \|v_1\|^2 = \int_{-1}^{1} x^2\,dx = \frac{2}{3},$$

so $(v_1, x^3)/\|v_1\|^2 = 3/5$. Therefore

$$v_3(x) = x^3 - \frac{3}{5}x, \qquad P_3(x) = \frac{v_3(x)}{v_3(1)} = \frac{1}{2}(5x^3 - 3x).$$

Getting $P_4(x)$ and beyond by Gram–Schmidt is possible, but recursion formulas—which the book develops later—are much faster.

Chapter summary

- Functions live in **inner product spaces**; orthogonality lets us read off expansion coefficients with a single integral, $a_j = (f, u_j)/\|u_j\|^2$.
- An operator is **self-adjoint** when $(u, \mathrm{L}v) = (\mathrm{L}u, v)$. Its eigenvalues are real and its eigenfunctions are orthogonal.
- A general second-order operator $p_0 u'' + p_1 u' + p_2 u$ is self-adjoint exactly when $p_0' = p_1$, in which case it is a **Sturm–Liouville operator** $\frac{d}{dx}(p\, u') + qu$.
- The associated **Sturm–Liouville equation** $(pu')' + qu = -\lambda wu$ has orthogonal eigenfunctions (weight w).
- A second-order operator with $p_0 \neq 0$ and integrable p_1/p_0 becomes Sturm–Liouville after multiplying by $Q = \exp\int(p_1/p_0)$. Legendre, Bessel, and Hermite all fit this mold.
- **Gram–Schmidt** on the monomials over $[-1, 1]$ generates the orthogonal **Legendre family**; the raw outputs $v_0 = 1,\ v_1 = x,\ v_2 = x^2 - \frac{1}{3}, \ldots$ become the standard P_n after scaling to $P_n(1) = 1$.